CARD 01 — PROPANE (C_3H_8)

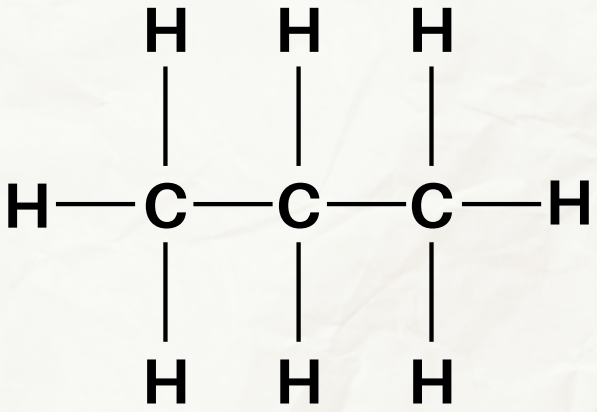
"THE SIMPLE SKELETON"

1. MOLECULAR FORMULA

$$C_3H_8$$

3 C + 8 H = 11 atoms

2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$CH_3CH_2CH_3$$

4. BOND-LINE FORMULA



CARBON TETRAVALENCY
"Each carbon forms four bonds, giving propane its stable three-carbon chain."

VERIFICATION STRIP

- ☑ atom count verified
- ☑ valence checked
- ☑ bond order checked
- ☑ same connectivity confirmed

ChE Connection: LPG and Fuel Processing

CARD 02 — ETHANOL (C_2H_6O)

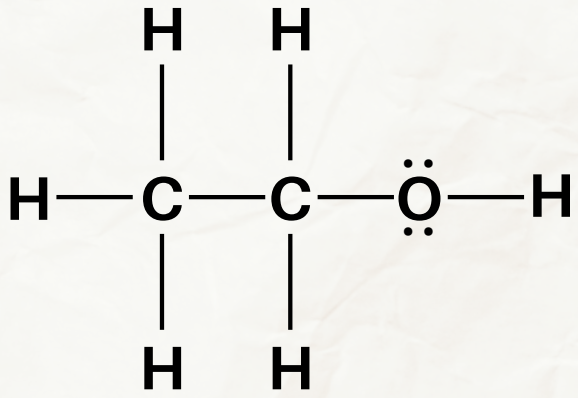
"FIRST HETEROATOM"

1. MOLECULAR FORMULA

$$C_2H_6O$$

2 C + 6 H + 1 O = 9 atoms

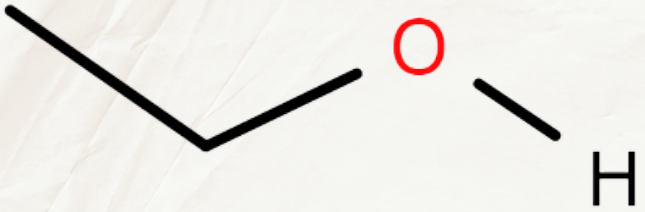
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$CH_3CH_2OH$$

4. BOND-LINE FORMULA



Ethanol features an oxygen atom that forms two single bonds and holds two lone pairs, while its -OH group signifies the alcohol functional group.

VERIFICATION STRIP

- ☑ atom count verified
- ☑ valence checked
- ☑ bond order checked
- ☑ same connectivity confirmed

ChE Connection: Fermentation, Distillation, Solvents, and Biofuels

CARD 03 — ACETONE (C_3H_6O)

(propan-2-one)

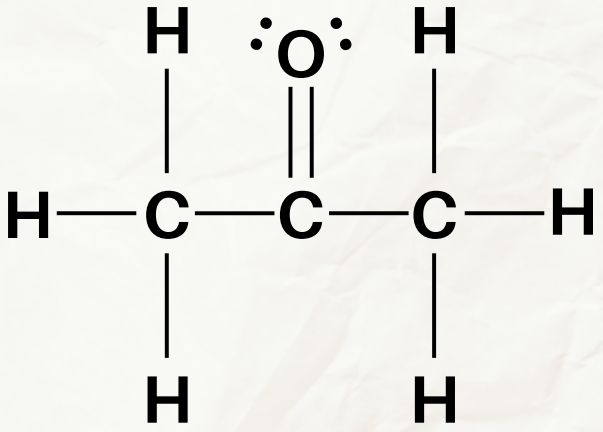
"THE CARBONYL CONNECTION"

1. MOLECULAR FORMULA

$$C_3H_6O$$

3 C + 6 H + 1 O = 10 atoms

2. LEWIS STRUCTURE

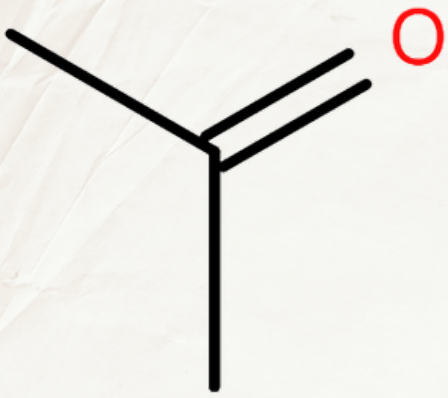


3. CONDENSED STRUCTURAL FORMULA

$$CH_3COCH_3$$

$CH_3C(=O)CH_3$

4. BOND-LINE FORMULA



This arrangement identifies acetone as a ketone and distinguishes its structure from ethanol, which contains a hydroxyl group (–OH).

VERIFICATION STRIP

- ☑ atom count verified
- ☑ bond order checked
- ☑ valence checked
- ☑ same connectivity confirmed

ChE Connection: Solvents, Chemical Processing, and Industrial Cleaning

CARD 04 — ACETIC ACID ($C_2H_4O_2$)

(ethanoic acid)

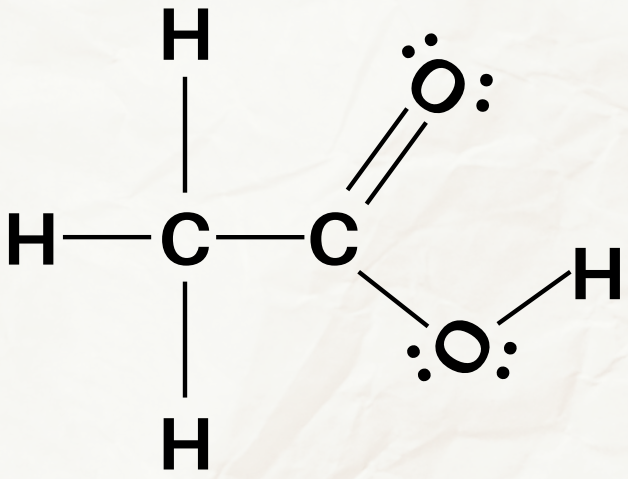
"THE ACIDIC O"

1. MOLECULAR FORMULA

$$C_2H_4O_2$$

2 C + 4 H + 2 O = 8 atoms

2. LEWIS STRUCTURE

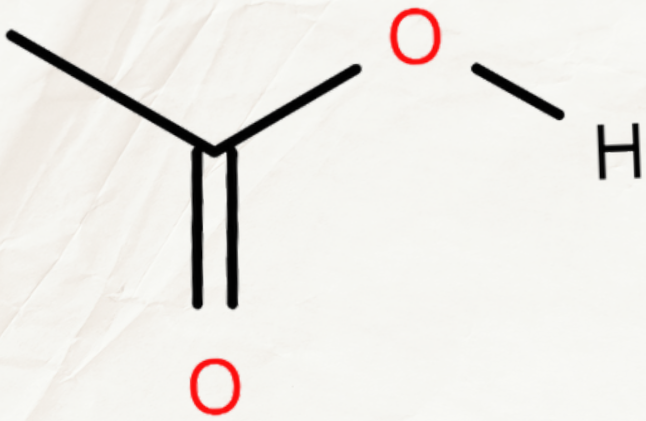


3. CONDENSED STRUCTURAL FORMULA

$$CH_3COOH$$

$CH_3C(=O)OH$

4. BOND-LINE FORMULA



Acetic acid contains a carboxyl group (-COOH), which combines a carbonyl group (C=O) and a hydroxyl group (-OH) on the same carbon atom.

VERIFICATION STRIP

- ☑ atom count verified
- ☑ bond order checked
- ☑ valence checked
- ☑ same connectivity confirmed

ChE Connection: Food Processing, Fermentation, and Chemical Production

CARD 05 — ETHYL ACETATE ($C_4H_8O_2$)

(ethyl ethanoate)

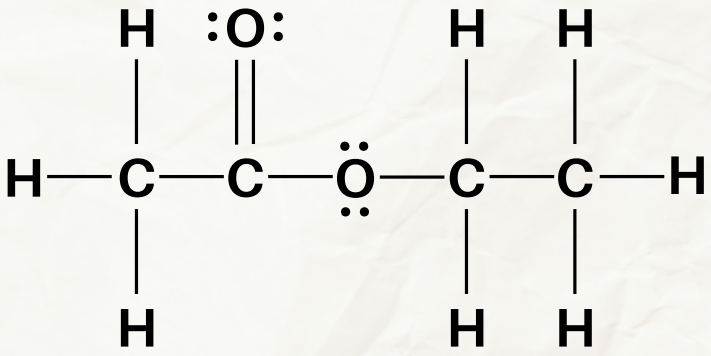
"THE ESTER CONNECTION"

1. MOLECULAR FORMULA

$$C_4H_8O_2$$

4 C + 8 H + 2 O = 14 atoms

2. LEWIS STRUCTURE

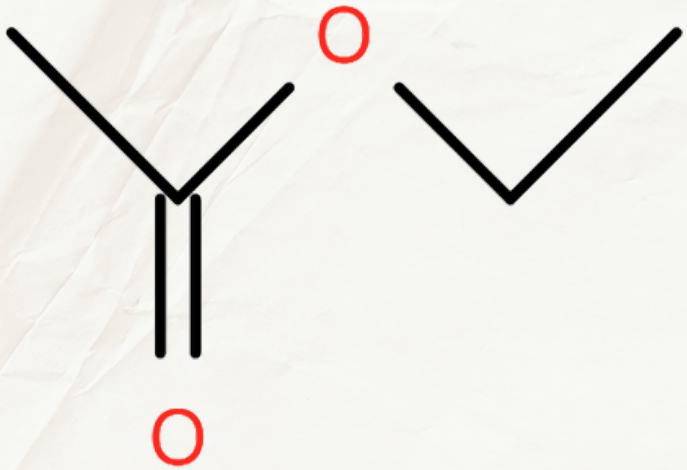


3. CONDENSED STRUCTURAL FORMULA

$$CH_3COOCH_2CH_3$$

$CH_3C(=O)OCH_2CH_3$

4. BOND-LINE FORMULA



Ethyl acetate contains the ester functional group. Esters have a carbonyl carbon bonded to another oxygen, which connects to a carbon group.

VERIFICATION STRIP

- ☑ atom count verified
- ☑ bond order checked
- ☑ valence checked
- ☑ same connectivity confirmed

ChE Connection: Solvent Extraction, Coatings, Inks, and Adhesives

CARD 06 — CYCLOHEXANE (C_6H_{12})

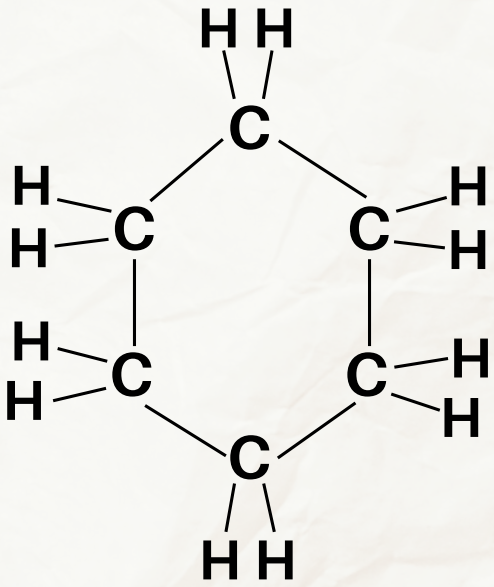
"THE FIRST RING"

1. MOLECULAR FORMULA

$$C_6H_{12}$$

6 C + 12 H = 18 atoms

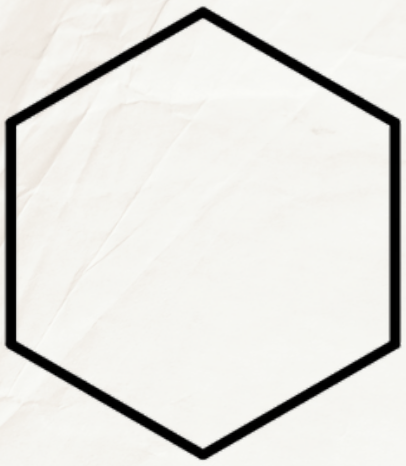
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$(CH_2)_6$$

4. BOND-LINE FORMULA



Cyclohexane is a cyclic alkane consisting of six carbon atoms linked in a ring with single bonds. Each carbon atom is bonded to two neighboring carbons and two hydrogen atoms.

VERIFICATION STRIP

- ☑ atom count verified
- ☑ valence checked
- ☑ bond order checked
- ☑ same connectivity confirmed

ChE Connection: Petroleum Processing, Nylon Production, and Industrial Feedstocks

CARD 07 — CYCLOHEXANONE ($C_6H_{10}O$)

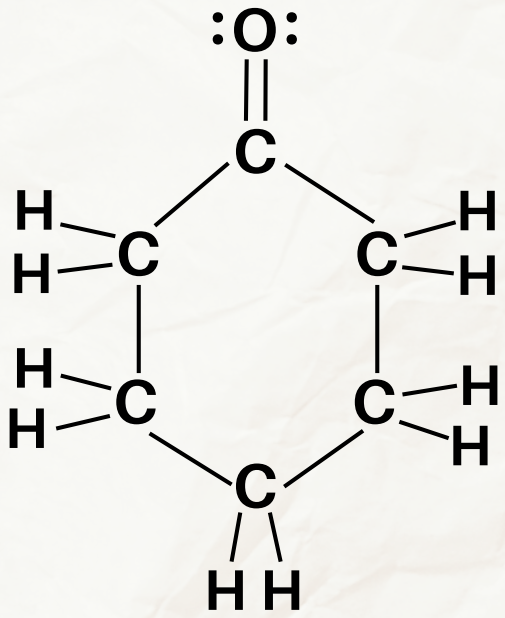
"THE CARBONYL IN A RING"

1. MOLECULAR FORMULA

$$C_6H_{10}O$$

6 C + 10 H + 1 O = 17 atoms

2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$(CH_2)_5CO$$

4. BOND-LINE FORMULA



Cyclohexanone is a six-carbon cyclic ketone with a carbonyl group formed by a double bond between one ring carbon and oxygen, distinguishing it from cyclohexane.

VERIFICATION STRIP

- ☑ atom count verified
- ☑ valence checked
- ☑ bond order checked
- ☑ same connectivity confirmed

ChE Connection: Nylon Intermediates, Solvents, and Chemical Manufacturing

CARD 08 — ACETAMINOPHEN ($C_8H_9NO_2$)

(paracetamol)

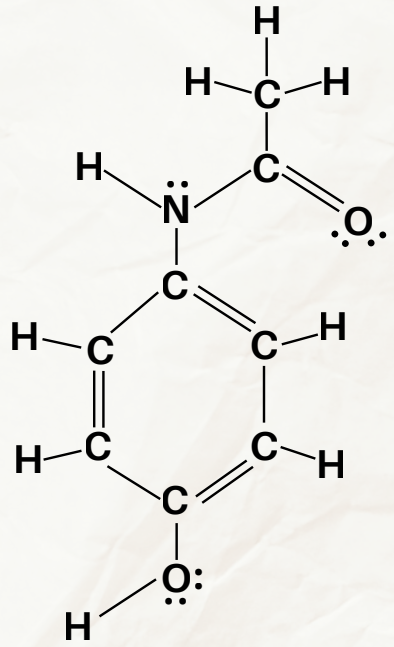
"THE AROMATIC AMIDE"

1. MOLECULAR FORMULA

$$C_8H_9NO_2$$

8 C + 9 H + 1 N + 2 O = 20 atoms

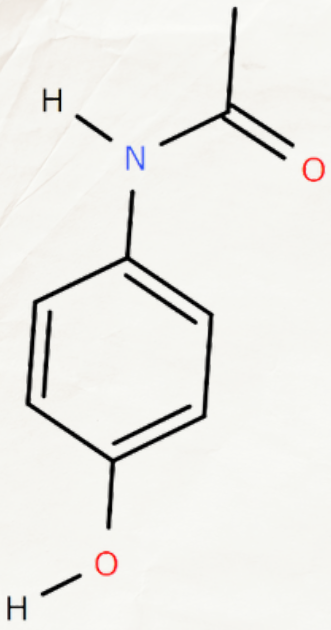
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$HO\text{-}C_6H_4\text{-}NHCOCH_3$$

4. BOND-LINE FORMULA



VERIFICATION STRIP

- ☑ atom count verified
- ☑ bond order checked
- ☑ valence checked
- ☑ same connectivity confirmed

ChE Connection: Pharmaceutical Manufacturing and Quality Control

CARD 09 — ASPIRIN ($C_9H_8O_4$)

(acetylsalicylic acid)

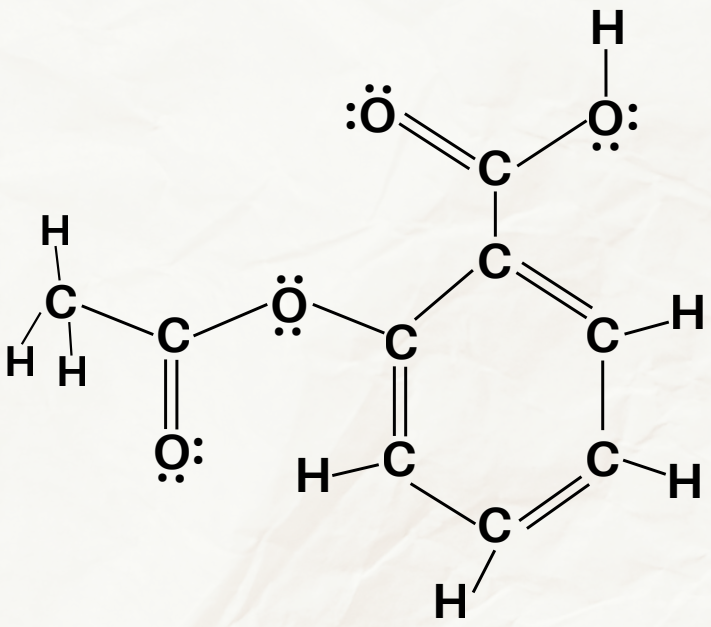
"TWO FUNCTIONAL GROUPS, ONE RING"

1. MOLECULAR FORMULA

$$C_9H_8O_4$$

9 C + 8 H + 4 O = 21 atoms

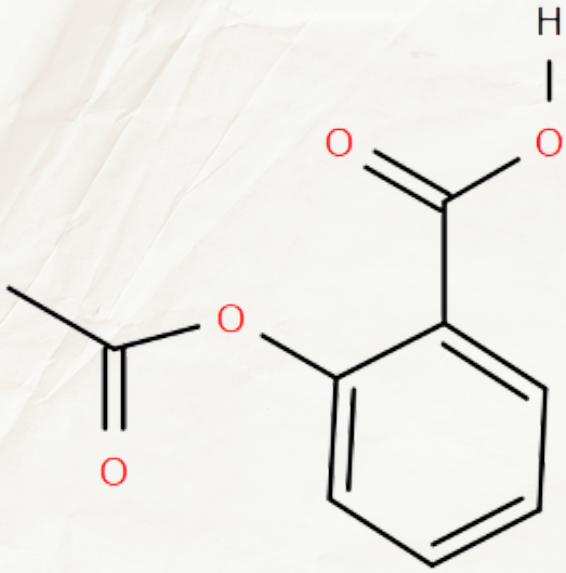
2. LEWIS STRUCTURE



3. CONDENSED STRUCTURAL FORMULA

$$C_6H_4(COOH)(OCOCH_3)$$

4. BOND-LINE FORMULA



VERIFICATION STRIP

- ☑ atom count verified
- ☑ bond order checked
- ☑ valence checked
- ☑ same connectivity confirmed

ChE Connection: Pharmaceutical Production, Chemical Synthesis, and Quality Control

CARD 10 — CAFFEINE ($C_8H_{10}N_4O_2$)

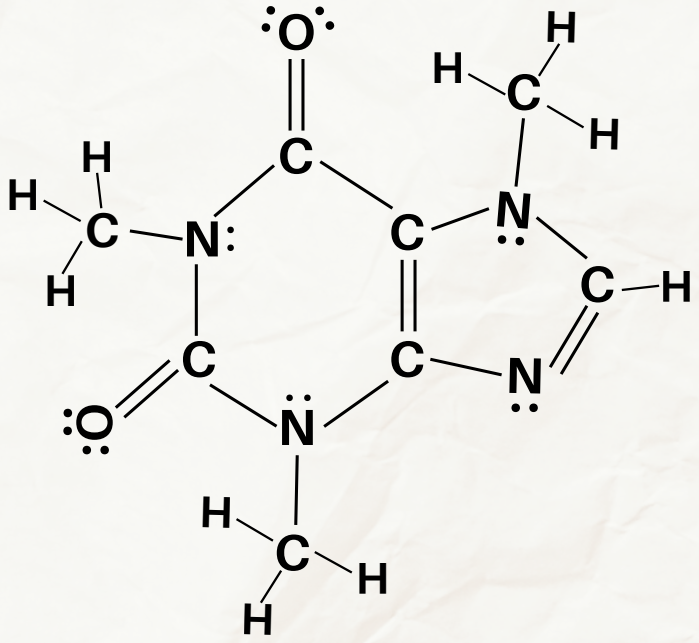
"THE FUSED-RING FINALE"

1. MOLECULAR FORMULA

$$C_8H_{10}N_4O_2$$

8 C + 10 H + 4 N + 2 O = 24 atoms

2. LEWIS STRUCTURE



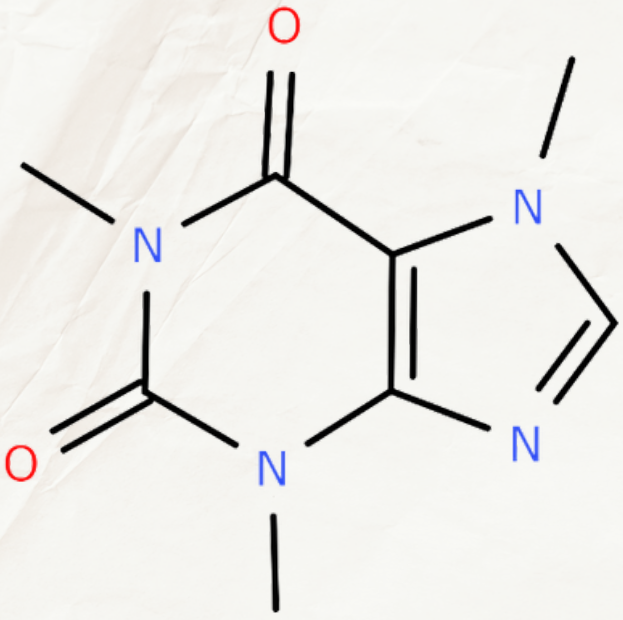
3. CONDENSED STRUCTURAL FORMULA

1,3,7-TRIMETHYLPURINE-2,6-DIONE

$CH_3N1(CO)N3(CH_3)C4{=}C5(N7(CH_3)C6{=}N9)CO$

Caffeine does not have a simple, universally used condensed structural formula comparable to CH_3CH_2OH.

4. BOND-LINE FORMULA



VERIFICATION STRIP

- ☑ atom count verified
- ☑ valence checked
- ☑ bond order checked
- ☑ same connectivity confirmed

ChE Connection: Food and Beverage Processing, Extraction, and Quality Control